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Consistency guided multiple plane image construction for novel view synthesis

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ABSTRACT

Recently, Multi-Plane Image shows great potential in novel view synthesis since it provide a generalizable formulation that enables strong reasoning ability even in unknown scenes. However, existing methods not only struggle with occlusion and complex scenes, but also often require a large number of depth planes and expensive computational cost. In this paper, we propose a novel consistency guided Multi-Plane Image construction method specifically designed for novel view synthesis. Different from the previous MPI-based methods, we construct the MPI serially layer by layer and accumulate consistency information during the process. Specifically we first propose a cross-view consistency mask to incorporate foreground occlusion information into the layered construction to perceive occlusion. Second, we propose a cross-layer consistency mask and a novel depth guidance strategy to incorporate appropriate scene context information into the layered construction to better understand the geometric structure of the scene. We conduct extensive experiments on the Spaces and Real Forward-Facing datasets. The results demonstrate that our method excels in novel view synthesis and multi-frame denoising tasks, achieving state-of-the-art performance with relatively low computational cost. Quantitatively, it outperforms state-of-the-art methods by improving PSNR by approximately 1.1% in challenging sparse-view settings and 2.1% in denoising tasks, achieving superior performance with relatively low computational cost.

1. Introduction

Novel view synthesis has long been a classic problem in computer graphics, with the aim of generating novel views of a scene based on one or more images captured from known camera positions. Currently, this technology has been widely applied in cutting-edge fields such as 3D reconstruction (Hu & Liu, 2025; Wang et al., 2021), interactive 3D photography (Shih et al., 2020) and virtual reality (Li et al., 2025b), with extremely broad application prospects.

In recent years, Neural Radiance Field (NeRF) (Barron et al., 2021; Mildenhall et al., 2020a; Wang et al., 2025) and 3D Gaussian (Huang et al., 2024a; Kerbl et al., 2023; Li et al., 2024) have demonstrated impressive rendering quality and real-time rendering speed in novel view synthesis. They are all based on neural representation and focus on the scene-specific formulation, where models are trained to fit one scene. Such scene-specific methods are often highly data-dependent and have poor generalization ability across different scenarios, which makes them face significant challenges in broader applications.

To maintain strong ability in reasoning about unknown scenes, some methods (Flynn et al., 2019; Kohler et al., 2023; Tanay et al., 2023) enable

able rendering novel views of unknown scenes directly from a set of posed input views via Multi-Plane Image (MPI). These methods represent the scene by predicting a stack of semi-transparent colored layers, consisting of a color map and an alpha map, which represent the color and transparency of the objects in the corresponding depth planes arranged at fixed depth range (Zhou et al., 2018).

Existing MPI-based methods mainly face the problems of occlusion and cross-layer inconsistency. First, to correctly represent the spatial regions that are occluded in some input views, the visibility difference between input views should be considered (Flynn et al., 2019). Failure to correctly represent these occluded regions will cause the occluded objects to disappear or become discontinuous in the synthesized view. Second, since the depth information of the scene in MPI is modeled through multiple discrete alpha maps, objects in different layers often appear inconsistent in complex scenes, which can result in surface discontinuities or artifacts at the edges of objects in the synthesized view (Tanay et al., 2023). As shown in Fig. 1, existing methods typically process the Plane Sweep Volume (PSV) to build MPI in parallel at each layer and try to solve the above problems by optimizing layer partitioning (Flynn et al., 2019) and a learned feed-forward network (Tanay et al., 2023). How-

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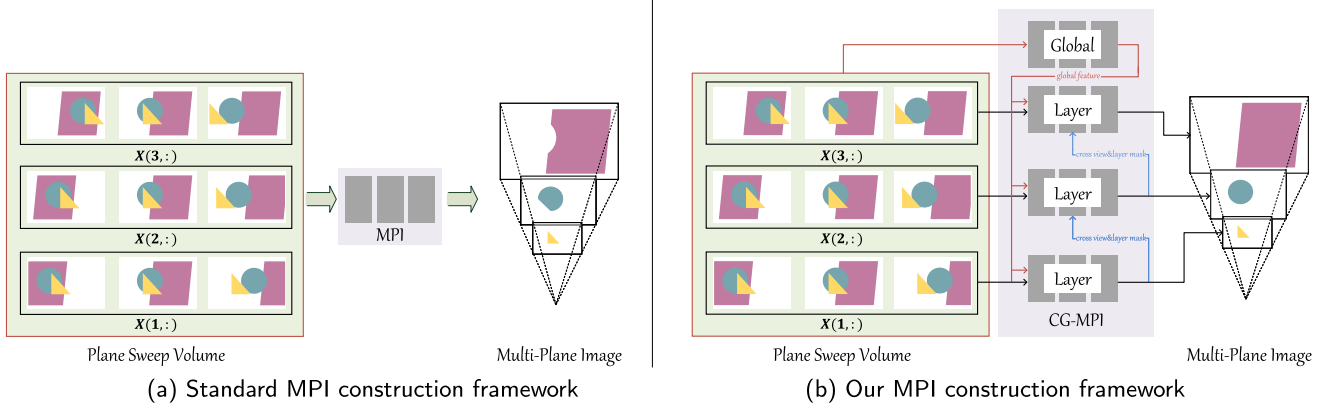


Fig. 1. Illustration of different frameworks for MPI construction. Left: Standard MPI construction methods generate all layers of MPIs simultaneously. Right: Our MPI construction method generates the layers of MPIs layer-by-layer in the order from near to far.

ever, the color and alpha predictions for each layer are independent and unknown to other layers, resulting in a large number of MPI layers and high computational cost to understand the occlusion relationships in the scene. In addition, this standard parallel depth processing method cannot effectively perceive scene context information and has difficulty maintaining consistency across layers.

As shown in Fig. 1 (b), we consider processing the PSV along the depth dimension MPI in a serial manner and accumulating foreground information progressively throughout the process. We observe that foreground content greatly affects the visibility and appearance of background depth layers. When foreground details are not adequately captured, accurately modeling occlusions and visibility differences cross views becomes difficult. By progressively gathering foreground information throughout the depth layers, each layer receives richer contextual cues, which allows for more precise visibility analysis and enhances the overall accuracy for occlusion handling. Meanwhile, to complement the detailed construction of each depth layer, we first perform a coarse-grained processing over all depths of the scene to gather global contextual feature. This global feature captures broader geometric structures and depth distribution that are difficult to infer from individual depth layers alone. Integrating this global context into the subsequent fine-grained MPI construction enables each depth layer to access information beyond its local neighborhood. This combination of global and local cues helps the model better understand the complex geometry of the scene and produce more coherent and consistent reconstruction.

Based on this understanding, we propose a Consistency Guided MPI construction framework (CG-MPI) for novel view synthesis. Specifically, we first propose a cross-view consistency mask to incorporate foreground occlusion information into the layered construction, effectively addressing the problem of the visibility between the predicted MPI and the input views. Second, we propose a cross-layer consistency mask and a novel global depth guidance strategy to incorporate appropriate scene context information into the layered construction, effectively addressing the problem of inconsistent object representations at different layers. The overall performance of our framework is substantially enhanced, achieving significant improvements over state-of-the-art methods in both novel view synthesis and multi-frame denoising. It also offers clear advantages in terms of computational efficiency and the number of depth layers required. Our main contributions are as follows:

1. We propose a cross-view consistency mask and a novel alpha blending strategy that can effectively guide the construction of MPI in occluded areas.
2. We propose a cross-layer consistency mask and a global depth guidance strategy that can efficiently ensure cross-depth consistency during the MPI construction process.

3. We propose a novel consistency-guided MPI construction method, which can effectively ensure the cross-view consistency and cross-layer consistency of MPI by considering the occlusion information of the foreground layer and aggregating the global feature of the scene.
4. Extensive experiments on two benchmark datasets show that our approach achieves superior performance compared to existing state-of-the-art methods in both novel view synthesis and multi-frame denoising.

2. Related works

2.1. Novel view synthesis

Recently, novel view synthesis has been approached as an encoding-rendering process where a scene representation is first encoded from a set of input images and an arbitrary number of novel views are then rendered from this scene representation, such as neural radiance field (NeRF) (Barron et al., 2021; Chen et al., 2021; Deng et al., 2022), Gaussians (Chung et al., 2024; Huang et al., 2024a; Kerbl et al., 2023; Li et al., 2025a), and Multi-Plane Image (MPI) (Zhou et al., 2018). NeRF and its variants (Barron et al., 2021; Mildenhall et al., 2020a; Wang et al., 2025) represent the scene implicitly as a continuous volume function through a multi-layer perceptron. These methods have received widespread attention for their ability to generate high-quality rendering of complex scenes from arbitrary views, while requiring significant computational resources, a large number of input views and separate training for each scene. Gaussian-based methods employ 3D Gaussian primitives to represent complex scenes, achieving real-time rendering of high-resolution images. However, popping and rendering artifacts are commonly present in Gaussian splatting (Radl et al., 2024), especially in occluded areas and weak-textured regions.

MPI-based methods employ a set of semi-transparent planes to represent the scene. The idea of using layered representations to model scenes can be traced back to the use of mattes and blue screens in special effects filmmaking (Smith & Blinn, 1996; Vlahos & Taylor, 1993), and was initially applied to view interpolation in Szeliski and Golland (1998). Compared to more complex modeling methods, this simple and efficient scene representation method is better suited for generating novel view images from a small set of forward-facing view images.

Some previous methods explicitly construct MPI using color and transparency maps, relying on fixed over-composite (Porter & Duff, 1984) for volumetric rendering, such as DeepView (Flynn et al., 2019), AdaMPI (Han et al., 2022). These methods offer clear structural design and good interpretability, with well-defined layer-wise decomposition that facilitates explicit modeling of geometry and consistent rendering. However, due to the limited expressiveness, they struggle to

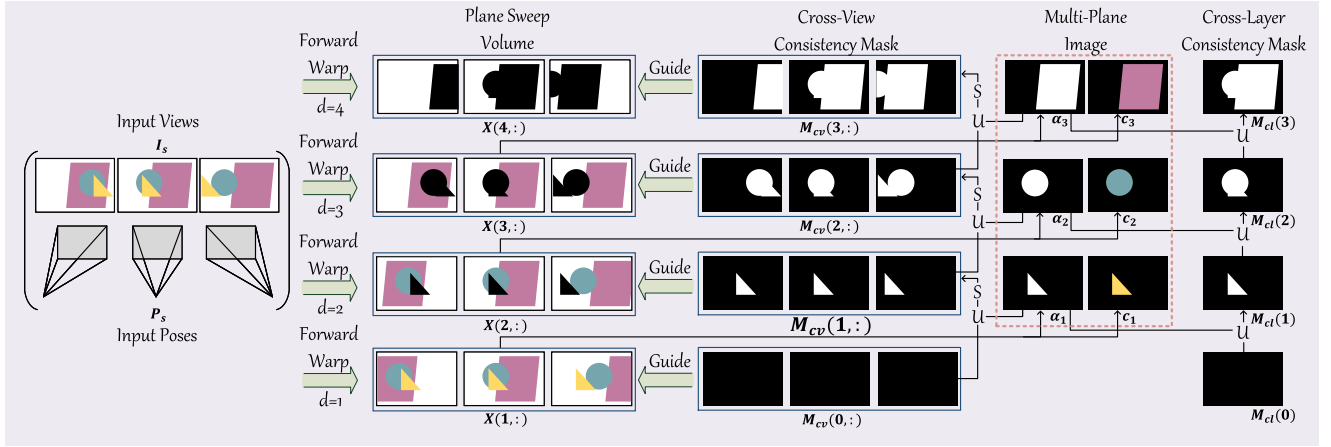


Fig. 2. Illustration of the calculation of the cross-view consistency mask and cross-layer consistency mask.

handle complex scenes with occlusions and fine-grained geometric details. Accurate representation of such details often requires robust edge perception (El-Sayed & Abd El-Hafeez, 2012) or advanced feature aggregation strategies (Eliwa & Abd El-Hafeez, 2025). Recent methods have proposed strategies to implicitly construct MPI. NeX (Phongthawee et al., 2023) has parameterized each pixel as a linear combination of basis functions with an MPI representation. MPFER (Tanay et al., 2023) replaces traditional RGB layers with high-dimensional feature maps and utilizes a learnable renderer instead of fixed compositing operations. ConvGLR (Tanay & Maggioni, 2024) proposes bypassing the MPI construction entirely, opting instead to directly render the PSV to the target view using a learnable renderer. By leveraging deep feature fusion and multi-view geometric cues, this approach achieves significant performance improvement. Although these methods significantly improve rendering quality, they often require a large number of depth layers and expensive computational cost. Moreover, these implicit modeling methods still face challenges when dealing with occlusion and complex scenes, especially when there are large changes in perspective, which may lead to incorrect rendering results.

2.2. Multi-frame denoising

Multi-frame denoising methods are generally classified into two categories based on the type of image alignment used. Explicit alignment refers to directly modeling inter-frame motion using optical flow or motion estimation modules (Caballero et al., 2017; Pan & Li, 2025; Tao et al., 2017; Tassano et al., 2019; Xue et al., 2019). For instance, VBM3D (Maggioni et al., 2012) and FastDVDnet (Tassano et al., 2019) use block matching or optical flow estimation to compensate for inter-frame motion. While explicit alignment can handle large-scale motions effectively, its performance is heavily dependent on the accuracy of optical flow estimation. On the other hand, implicit alignment learns alignment indirectly through the internal mechanism of the model. For example, TDAN (Tian et al., 2020) and EDVR (Wang et al., 2019) utilize deformable convolutions to achieve implicit alignment in the feature space. Implicit alignment is typically more robust for handling small-range motions but tends to struggle in scenarios involving large motions. Recent studies (Chan et al., 2022) have introduced hybrid alignment in the form of flow-guided deformable convolutions, which can deliver state-of-the-art performance.

Image restoration tasks, ranging from super-resolution (Hassan et al., 2025) to multi-frame denoising, share the common goal of recovering high-fidelity content from degraded inputs. So far, the application of view synthesis methods in multi-frame denoising has been relatively limited, and most are based on NeRF (Mildenhall et al., 2020a). RawN-

eRF (Mildenhall et al., 2022) explores novel view synthesis under low-light conditions and demonstrates significant denoising effects in the synthesized views; Deblur-NeRF (Ma et al., 2022) enhances NeRF's ability to handle blurriness by introducing deformable sparse kernels; Noiseaware-NeRFs (Pearl et al., 2022) improves the IBRNet architecture to explicitly handle noise. However, these different restoration methods are still limited by the inherent flaws of their underlying NeRF representations. MPFER first explores the application of MPI-based novel view synthesis methods in multi-frame denoising and achieves favorable denoising results.

3. Consistency mask calculation

The MPI construction pipeline (Flynn et al., 2019; Han et al., 2022; Tanay et al., 2023) takes a set of views $I_s \in \mathbb{R}^{V \times H \times W \times 3}$ and the corresponding pose parameters P_s as input to produce multiple semi-transparent layers. Each layer consists of a color map $c \in \mathbb{R}^{H \times W \times 3}$ and an alpha map $\alpha \in \mathbb{R}^{H \times W \times 1}$, which respectively represent the color and transparency of the objects within the corresponding depth layer (Zhou et al., 2018).

3.1. Standard MPI construction

The MPI construction pipeline mainly consists of two steps: PSV calculation and MPI prediction.

3.1.1. PSV calculation

PSV is a 5D tensor $\mathcal{X} \in \mathbb{R}^{D \times V \times 3 \times H \times W}$, obtained by forward-warping each input image on each of the D depth layers parallel to a reference view $I_* \in \mathbb{R}^{H \times W \times 3}$:

$$\mathcal{X} = \mathcal{W}(I_s, \{J_i\}_{i=1}^D) \quad (1)$$

where $\mathcal{W}$ denotes the warping operation and $\{J_i\}_{i=1}^D$ are homography matrices induced by the depth planes.

Each homography matrix is determined by the pose parameters P_s of the input views, consisting of an intrinsic tensor $K_s \in \mathbb{R}^{3 \times 3 \times 3}$ and an extrinsic tensor that can be split into a rotation tensor $R_s \in \mathbb{R}^{V \times 3 \times 3}$ and a translation tensor $t_s \in \mathbb{R}^{V \times 3 \times 1}$. For the reference view I_* , we denote its pose as $P_* = \{K_*, R_*, t_*\}$. The D depth layers are defined by distances $\{d_i\}_{i=1}^D$ from the reference camera center (we discuss the choice of these distances in practice in Section 5.2), sharing a common normal n_* . Assuming the world origin at the reference camera center ($R_* = I$, $t_* = 0$, $n_* = [0, 0, 1]^T$), the homography from an input view to the reference view at depth d_i is given by Hartley and Zisserman (2003):

$$J_i = K_* \left(R_* - \frac{t_* t_*^T}{d_i} \right) K_s^{-1}. \quad (2)$$

Forward-warping all input images via these homographies yields the PSV, which encodes multi-view geometry across depth planes.

The PSV is a highly structured tensor that encodes the epipolar geometry between the input images and the reference images (Szeliski & Golland, 1998). For a given pixel location $[h, w]$ in the reference image plane, its corresponding camera ray projects as a set of epipolar lines in the input images. By sampling along these epipolar lines at matching depths, the PSV slice: $\mathcal{X}_{hw} = \mathcal{X}[:, :, :, h, w]$ captures the consistency and geometric correspondences across images. $\mathcal{X}_{hw}$ aggregates image features from different views and depths at this spatial location, effectively encoding the camera ray passing through $[h, w]$. Moreover, due to its well-structured arrangement along the view, depth and spatial dimensions, the PSV enables efficient spatial geometry modeling and feature fusion operations. For instance, aggregation along the view dimension extracts depth-related features pointwise, while operations along the depth dimension model the variances between different views.

3.1.2. MPI prediction

As shown in Fig. 1 (a), the standard MPI construction pipeline usually generate all layers of MPIs simultaneously, the alpha of each layer is generated independently. Although it is intuitive and simple, it does not specifically consider the occlusion relationship of the scene. Without explicitly considering the occlusion relationship between different layers, the occluded parts that need to recover cannot be well modeled in the specific layers. For example, parts of the circle and parallelogram in Fig. 1 (a) are occluded by the foreground, and these occluded areas destroy the inter-view consistency. If the foreground occlusion is not specifically considered when constructing the current layer, it is difficult to fully restore the objects at the current depth in the MPI. For example, in the MPI constructed in Fig. 1 (a), except for the nearest triangle, objects at other depths are partially missing.

Furthermore, the standard MPI construction pipeline often fuses PSV features along the depth dimension. While this approach helps focus on inter-view information fusion during MPI construction, it loses the context of the depth spaces within the PSV, destroying the inter-depth consistency. Without specifically incorporating depth context during MPI construction, the depth of some pixels cannot be accurately classified, resulting in artifacts, especially in areas with large depth variations, such as object edges.

3.2. Occlusion perception based on cross-view consistency

To reduce the interference from foreground occlusions, we propose a cross-view consistency mask to guide MPI construction, which is defined as $\mathcal{M}_{cv} \in \mathbb{R}^{D \times V \times 1 \times H \times W}$ to illustrate whether a pixel belongs to occlusion in each input view, and updated during the layer-by-layer construction of MPI. This mask indicates the position of foreground occlusions so that when generating the following layers, our model can focus on the regions not occluded by foreground. Specifically, as shown in Fig. 2, since there are no foreground objects in front of the 1-st (nearest) layer, the corresponding mask $\mathcal{M}_{cv}[0, :]$ is initialized to zeros. For the i th layer of MPI ($i \geq 1$), the objects indicated in the current alpha map are occlusions of the next layer, so we use the current alpha map α_i to update the corresponding cross-view consistency mask $\mathcal{M}_{cv}[i-1, :]$. α_i is expanded to a 4D tensor ($V \times 1 \times H \times W$) by repeating in the view dimensions and cumulated to $\mathcal{M}_{cv}[i-1, :]$, which we defined as $\mathcal{U}$ in Fig. 2:

$$\mathcal{U}(\mathcal{M}_{cv}[i-1, :], \alpha^i) = 1 - (1 - \mathcal{M}_{cv}[i-1, :]) \cdot (1 - \alpha^i) \quad (3)$$

Since the position information of the foreground objects before the i th layer is already contained in $\mathcal{M}_{cv}[i-1, :]$, $\mathcal{M}_{cv}(i, :)$ contains the position information of all foreground objects in front of the $(i+1)$ th layer. Finally, we use the product of the homography matrix $\mathcal{J}_i$ and the inverse homography matrix $\mathcal{J}_{i+1}^{-1}$ to layered shift $\mathcal{M}_{cv}(i, :)$ onto the next depth plane to align it with the foreground object in the $\mathcal{X}(i+1, :)$ by, which

we define as S in Fig. 2:

$$\mathcal{M}_{cv}[i, :] = S(\mathcal{U}(\mathcal{M}_{cv}[i-1, :], \alpha^i), \mathcal{J}_i \cdot \mathcal{J}_{i+1}^{-1}) \quad (4)$$

As shown in Fig. 2, after obtaining the cross-view consistency mask required for the next layer, we apply it to $\mathcal{X}[i+1, :]$ to guide the generation of the c_{i+1} and α_{i+1} . Since $\mathcal{M}_{cv}[i, :]$ explicitly provides the occluded regions for each input view, the model can clearly perceive the visibility differences between views and fully recover objects at current depth. Iteratively, the color and alpha maps of all layers are fully constructed from near to far, without being restricted by depth distance. As shown in Fig. 2, compared with the traditional MPI construction pipeline, the MPI constructed by our method has a more complete structure and is not affected by occlusion.

3.3. Depth perception based on cross-layer consistency

Furthermore, to process PSV along the depth dimension without losing the contextual information of the PSV depth space, we propose a global depth guidance strategy and a cross-layer consistency mask to guide MPI construction. Specifically, we encode the PSV as a global feature shared by each layer to incorporate information about the geometric structure of the scene in layered construction. This approach does not require a large number of depth planes or additional processing of PSV along the view dimension, avoiding high computational costs, thus making it easier to train and generalize to test data. However, if handled naively, the global feature provides independent guidance for each layer and is agnostic to other layers. Therefore, we further propose a cross-layer consistency mask, which is defined as $\mathcal{M}_{cl} \in \mathbb{R}^{D \times 1 \times H \times W}$ to illustrate whether a pixel belongs to the foreground in the reference view. This mask can filter the depth information contained in the global feature, allowing each plane to share the global feature while focusing on the required depth range information according to its own depth, thereby reducing the interference from redundant information. Specifically, as shown in Fig. 2, we initialize $\mathcal{M}_{cl}[0]$ to zeros and generate iteratively through operation $\mathcal{U}$ for the i th layer of MPI ($i \geq 1$) (Section 3.2).

$$\begin{aligned} \mathcal{M}_{cl}[i] &= \mathcal{U}(\mathcal{M}_{cl}[i-1], \alpha^{i-1}) \\ &= 1 - (1 - \mathcal{M}_{cl}[i-1]) \times (1 - \alpha^{i-1}) \end{aligned} \quad (5)$$

This entire process can transmit depth information only through multiple inter-layer iterations. It enables information interaction across the full depth range without relying on a large number of depth layers or complex network structures, thus maintaining computational efficiency while effectively preserving inter-layer consistency, allowing the network to recover the depth details of the scene more accurately. It is worth noting that $\mathcal{M}_{cv}$ is aligned with each input view, while $\mathcal{M}_{cl}$ is only aligned with the reference view. When the reference view is included in the input view, assuming the reference view is r , $\mathcal{M}_{cl}$ is equal to $\mathcal{M}_{cv}[:, r]$.

4. Consistency guided multiple plane image

The framework of our method is shown in Fig. 3. We first forward-warp the input views I_s using the homography matrix $\{\mathcal{J}\}_{i=1}^D$ according to the depth range of the scene to obtain the PSV $\mathcal{X}$ according to Eq. (1). Then we construct the MPI with the proposed consistency guided layer calculation. Finally, the target views $I_t \in \mathbb{R}^{V' \times H \times W \times 3}$ are generated by backward-warping and MPI rendering according to the corresponding pose parameters P_t .

$$I_t = H(I_s, P_s, P_t) \quad (6)$$

4.1. CG-MPI layer calculation

In Section 3.2, we propose a cross-view consistency mask that explicitly contains the occluded regions for each input view. To effectively

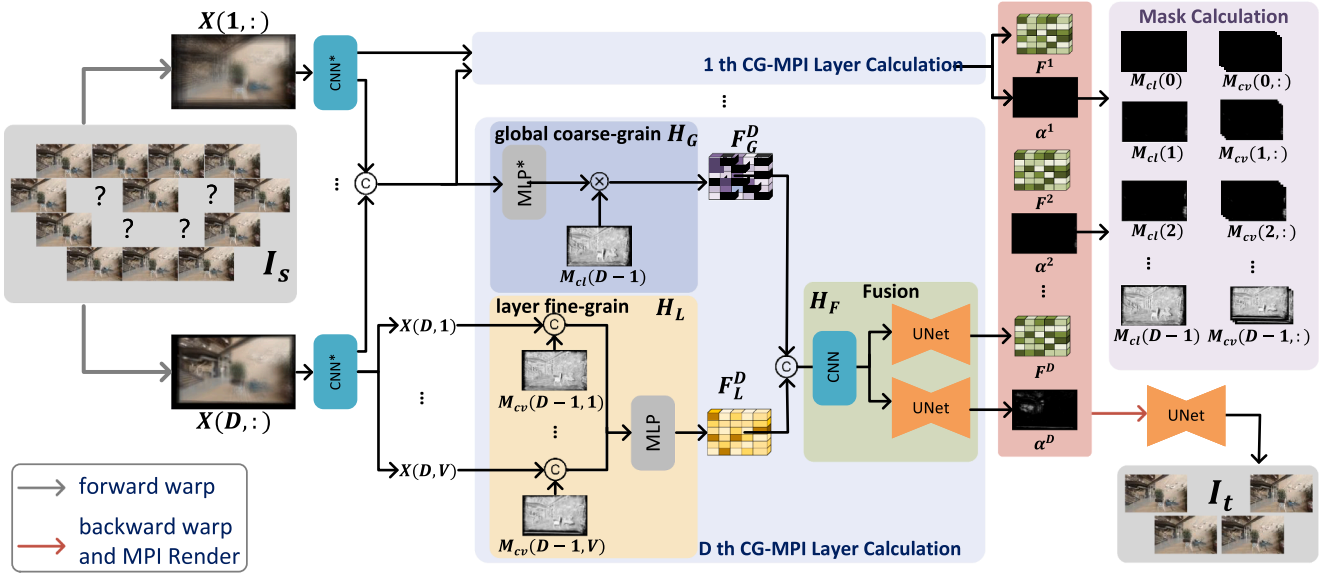


Fig. 3. The pipeline of the proposed consistency guided multi-plane images construction for novel view synthesis. * represents parameter sharing.

integrate this prior, we design the layer fine-grained module $\mathcal{H}_L$ as a pixel-wise Multi-Layer Perceptron network. Specifically, as shown in Fig. 3, for the D th layer, we concatenate $\mathcal{X}(D, :)$ with its corresponding cross-view consistency mask $\mathcal{M}_{cv}[D, :]$ along the view dimension. This results in a comprehensive feature vector that encapsulates both texture information and visibility cues for all views at the current depth. The MLP then projects this high dimensional view-aggregated feature into a compact latent embedding. By prioritizing high visibility regions guided by $\mathcal{M}_{cv}$, this pixel-wise processing allows the model to selectively fuse valid textures from visible views to hallucinate occluded regions, yielding the layered feature $\mathcal{F}_L^D$:

$$\mathcal{F}_L^D = \mathcal{H}_L(\{\mathcal{X}[D, j], \mathcal{M}_{cv}[D, j]\}_{j=1}^V) \quad (7)$$

As explained in Section 3.3, if the PSV is processed only along the depth dimension through multiple layered encoders, it will cause a serious loss of PSV depth space context information. To address this, we introduce a global encoder $\mathcal{H}_G$ to extract scene-level context. Instead of standard spatial convolutions, we formulate $\mathcal{H}_G$ to operate on the depth-view joint domain. Specifically, as shown in Fig. 3, we re-slice the PSV slices at all depths and input them into the multi-layer perceptron in $\mathcal{H}_G$ to fuse the information from the depth dimension and the view dimension simultaneously. Since this fusion process is only a coarse-grained processing, the fused features contain a large amount of redundant information for a specific depth layer, which will seriously affect the role of effective information. Therefore, we use the proposed $\mathcal{M}_{cl}$ (Section 3.3) to filter the fused features. As shown in Fig. 3, taking the D th layer as an example, we multiply $\mathcal{M}_{cl}[D-1]$ with the fused features, thereby masking the foreground information that has been considered in the calculation of the previous layers, and finally generate a global feature $\mathcal{F}_G^D$ for the current depth.

$$\mathcal{F}_G^D = \mathcal{H}_G(\mathcal{X}, \mathcal{M}_{cl}[D]) \quad (8)$$

To fully integrate the global geometric information and contextual semantics embedded in the global feature into the layered features, we first concatenate the refined global feature $\mathcal{F}_G^D$ with the local layered feature $\mathcal{F}_L^D$ along the channel dimension. The concatenated features are then fed into our designed fusion module $\mathcal{H}_F$ consisting of multiple convolutional layers to deeply fuse the two types of features. Unlike the previous pixel-wise modules, $\mathcal{H}_F$ employs 2D convolutional layers to aggregate spatial neighborhood information, en-

hancing the modeling of local geometric structures. The fused features are subsequently fed into two structurally identical but parameter-independent U-Net (Ronneberger et al., 2015). These two branches are responsible for predicting the feature map and the transparency map, respectively.

$$(\mathcal{F}^D, \alpha^D) = \mathcal{H}_F(\mathcal{F}_G^D, \mathcal{F}_L^D) \quad (9)$$

By iterating and performing our layered calculation starting from the first layer, we can obtain the complete MPI structure $\{(\mathcal{F}^i, \alpha^i)\}_{i=1}^D$.

4.2. Backward-warp and MPI rendering

After the MPI are constructed, we first apply backward-warping to project the complete set of $\{(\mathcal{F}^i, \alpha^i)\}_{i=1}^D$ into V' novel views using a set of homography matrices $\mathcal{J}_i$ based on the pose parameters P_* and P_i : $\mathcal{J}_i = \{K_i(R_i - \frac{t_i t_i^T}{d_i})K_*^{-1}\}_{i=1}^D$.

$$\{(\mathcal{F}_v^i, \alpha_v^i)\}_{i=1}^D \big|_{v=1}^{V'} = \mathcal{W}^{-1}(\{(\mathcal{F}^i, \alpha^i)\}_{i=1}^D, \mathcal{J}_i) \quad (10)$$

Then, we apply a classical front-to-back compositing operator to fuse the multi-plane features. This compositing process follows a depth ordering from foreground to background and uses the alpha maps to effectively control the contribution of each layer.

$$\{\mathcal{F}_v\}_{v=1}^{V'} = \left\{ \frac{\sum_{i \in D} \mathcal{F}_v^i \cdot \alpha_v^i \prod_{i > j} (1 - \alpha_v^j)}{\sum_{i \in D} \alpha_v^i \prod_{i > j} (1 - \alpha_v^j)} \right\}_{v=1}^{V'} \quad (11)$$

where $\mathcal{F}_v^i$ and α_v^i are the backward-warped results of aligning $\mathcal{F}^i$ and α^i with view v . The cumulative amount of alpha maps from the remaining layers (the denominator portion) is used to correct the luminous flux.

Then we input the $\{\mathcal{F}_v\}_{v=1}^{V'}$ into the UNet, which we defined as $\mathcal{H}_u$ and output the image from the novel view.

$$I_t = \mathcal{H}_u(\{\mathcal{F}_v\}_{v=1}^{V'}) \quad (12)$$

Our approach is summarized in Algorithm 1.

4.3. Loss function

The loss function of our model is the L1 loss computed between the model's final output I_t and the ground truth image I_{gt} . By directly comparing the generated image to the reference image at the pixel level, this

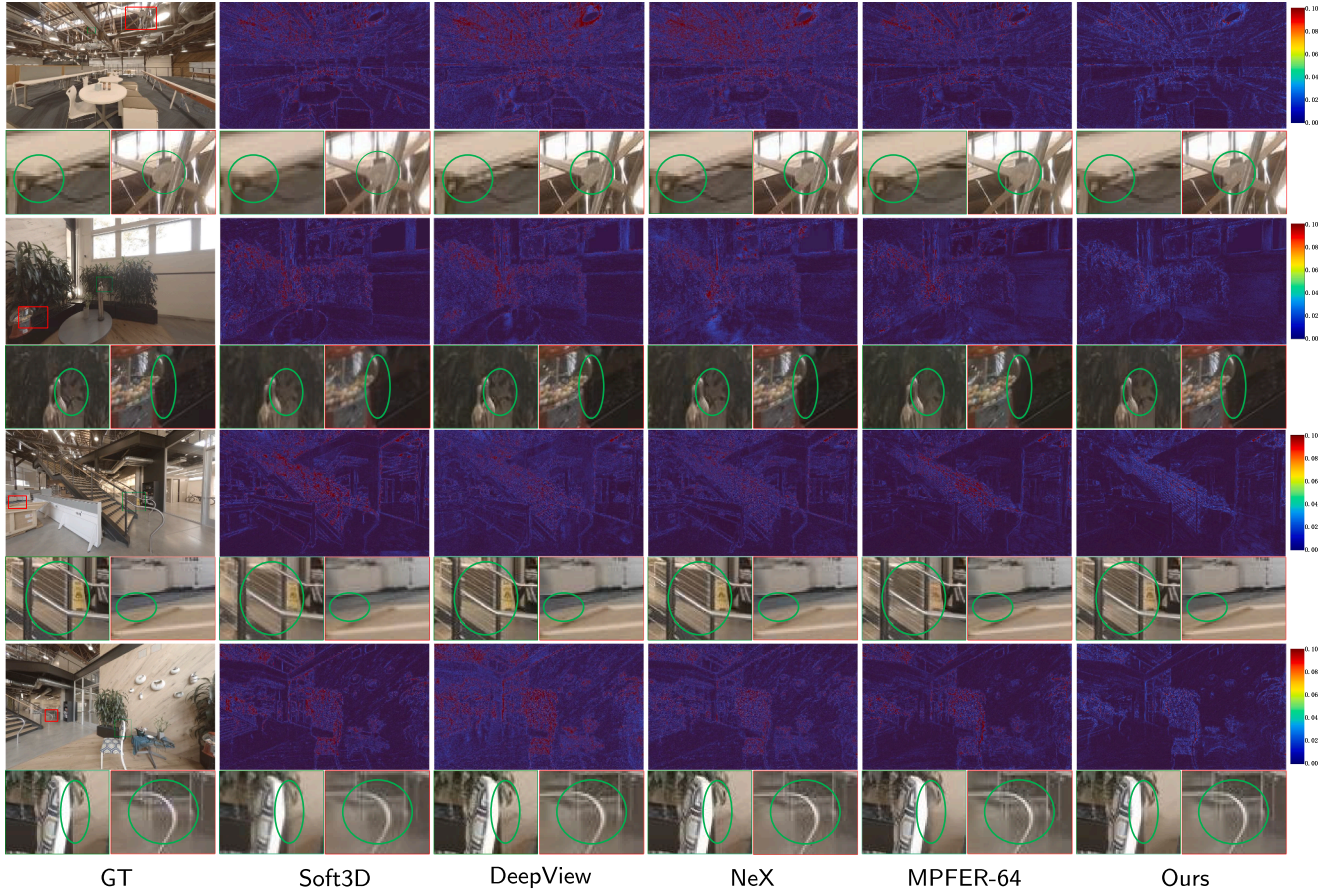


Fig. 4. Qualitative comparisons under the setup of 12 input views on the Spaces Dataset (Flynn et al., 2019).

Table 1

Quantitative comparisons of the setups with 12 input views and 4 input views on the space dataset. $\uparrow$ means the bigger the better. The best PSNR/SSIM result is highlighted in red and the second best result is highlighted in blue.

Method	D	12 input views		4 input views (small)		4 input views (medium)		4 input views (large)	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Soft3D (Penner & Zhang, 2017)	–	31.93	0.940	30.29	0.925	30.84	0.930	30.57	0.931
DeepView (Flynn et al., 2019)	80	34.23	0.965	31.42	0.954	32.38	0.957	31.00	0.952
NeX (Phongthawee et al., 2023)	192	35.84	0.985	–	–	–	–	–	–
SIMPLI (Solovev et al., 2022)	8	–	–	–	–	–	–	30.47	0.923
fMPI-L (Kohler et al., 2023)	40	–	–	–	–	–	–	33.69	0.966
MPFER (Tanay et al., 2023)	64	35.73	0.972	33.20	0.959	33.47	0.959	32.38	0.953
ConvGLR (Tanay & Maggioni, 2024)	128	36.05	0.977	<u>34.07</u>	0.968	<u>34.33</u>	0.968	33.34	0.964
Ours-S	16	<u>36.08</u>	<u>0.986</u>	33.55	<u>0.978</u>	33.55	<u>0.978</u>	32.21	<u>0.972</u>
Ours-L	48	36.29	0.987	34.15	0.981	34.53	0.982	33.70	0.979

loss supervises the quality of the final rendered results. It ensures that during each training iteration, the renderer parameters are adjusted so that the synthesized image more closely matches the visual appearance of the real scene. The objective function is defined as:

$$\mathcal{L} = \left\| I_t - I_{gt} \right\|_1 \quad (13)$$

5. Experiments

In this section, we first provide a detailed introduction to the datasets used in our experiments. Then we present the specific implementation details and a comprehensive description of the compared methods. Finally, we thoroughly present and analyze all experimental results.

5.1. Datasets

To comprehensively evaluate the effectiveness of our method, we adopted two publicly available datasets: DeepView (Flynn et al., 2019) and RFF (Mildenhall et al., 2020b).

1. Spaces Dataset: This dataset contains 100 indoor and outdoor scenes. Each scene is captured 5 to 14 times by 16 cameras from different views, with each image having a resolution of 480×800 . We use 90 of these scenes for training and the remaining 10 for evaluation.
2. RFF Dataset: This dataset contains 48 indoor and outdoor scenes, with each scene including 20 to 62 images captured by a handheld camera. The camera poses are computed via COLMAP. We use 40 of these scenes for training and the remaining 8 for evaluation.

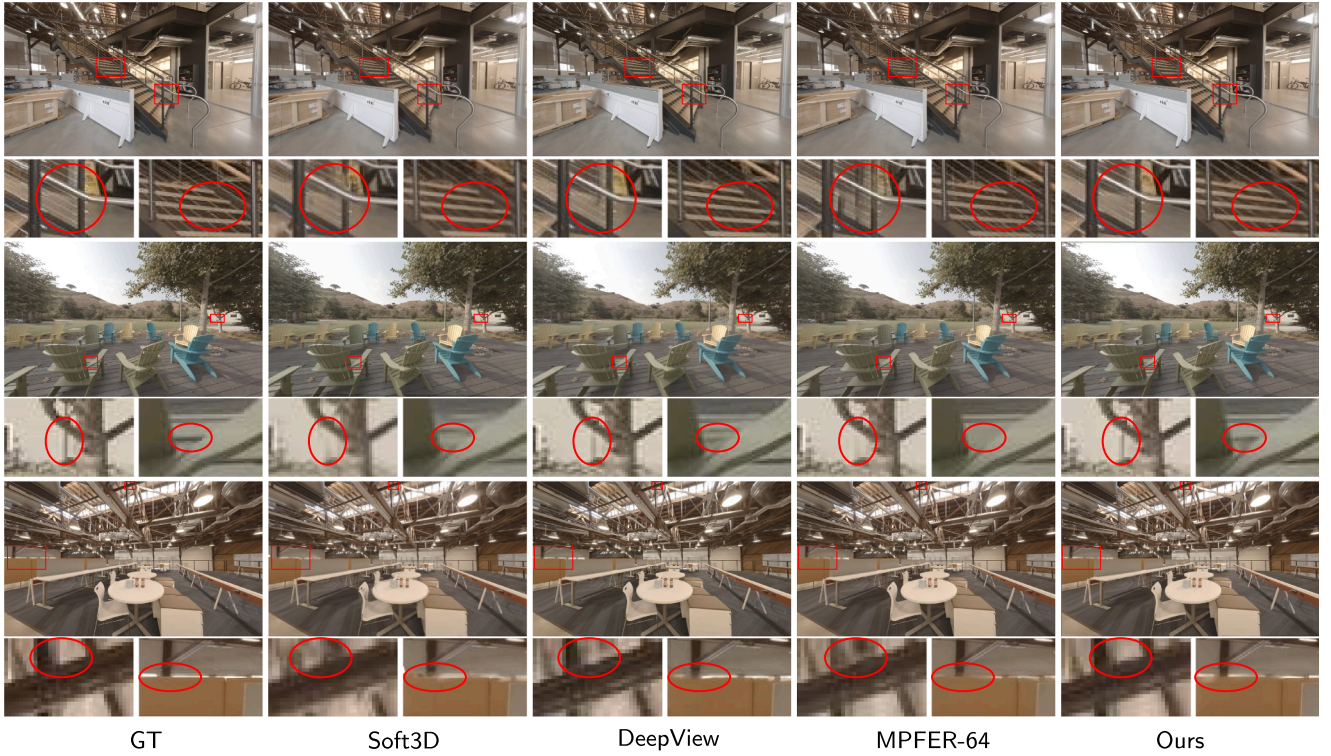


Fig. 5. Qualitative comparisons under the large baseline setup of 4 input views on the Spaces Dataset (Flynn et al., 2019).

Algorithm 1 CG-MPI inference: Serial construction and rendering.

Input: Source features $\{X_d\}_{d=1}^D$, Homography matrices $\{J_d\}_{d=1}^{D-1}$, Networks H_e, H_m, H_u ;

Initialize: Accumulation masks $\mathcal{M}_{cv} \leftarrow 0, \mathcal{M}_{cl} \leftarrow 0$;

Stage 1: Global Context Aggregation

$\mathcal{G} \leftarrow H_m(\{X_d\})$;

Stage 2: Serial MPI Construction

for $d = 1$ to D **do**

1. *Plane Encoding*:

$(F_d, \alpha_d) \leftarrow H_e(X_d, \mathcal{G}, \mathcal{M}_{sum}, \mathcal{M}_c)$;

Store MPI plane $\mathcal{P}_d \leftarrow \{F_d, \alpha_d\}$;

2. *Mask Accumulation*:

$\mathcal{M}_{cl} \leftarrow 1 - (1 - \mathcal{M}_{cl}) \odot (1 - \alpha_d)$;

$\mathcal{M}_{cv} \leftarrow 1 - (1 - \mathcal{M}_{cv}) \odot (1 - \alpha_d)$;

3. *Layer Shift*:

$\mathcal{M}_{cv} \leftarrow \text{Warp}(\mathcal{M}_{cv}, J_{d \rightarrow d+1})$;

end for

Stage 3: Target View Rendering

$\{\tilde{\mathcal{P}}\} \leftarrow \text{Backward-Warp}(\{\mathcal{P}\}, \text{Target Poses})$;

$\hat{I}_{tgt} \leftarrow H_u(\{\tilde{\mathcal{P}}\})$;

Output: Target image $\hat{I}_{tgt}$;

5.2. Implementation and training details

Our framework is implemented in PyTorch and trained using a single NVIDIA RTX A4000. This software environment relies entirely on open-source libraries, requiring no additional licensing fees. This configuration ensures the ease of use of the proposed method and allows deployment on standard scientific hardware without the need for expensive computing clusters. We verified the effectiveness of the model on the view synthesis task and the denoising task respectively.

View Synthesis Task: Following representative research works, we train and test our method on the Spaces dataset for the novel view synthesis task with fixed input and output view positions using four experimental settings: 12 input views and 4 output views; 4 input views and 3 output views (small baseline); 4 input views and 6 output views (medium baseline); and 4 input views and 9 output views (large baseline).

To validate the flexibility of our method, we also conduct a novel view synthesis task on the Spaces dataset with variable input and output view positions. This involves randomly selecting 15 of the 16 views as input views and the remaining 1 view as output. Furthermore, to verify the generalization of our method, we train an additional model on the Spaces dataset, randomly selecting 3 views as input and all remaining views as output, and test it on the RFF dataset.

To compare the impact of the number of depth planes on model performance, we implemented two settings, Ours-S and Ours-L, with 16 and 48 depth planes respectively. All depth planes are distributed between 100 m and 0.5 m from the reference camera, and we selected the central view as the position of the reference camera.

Denoising Task: To verify the performance of our method in the multi-frame denoising task, similar to previous studies (Tanay et al., 2023), the input consists of 16 view images from one rig position with images degraded by noise, and the targets are the same 16 views denoised. We apply synthetic noise with a signal dependent gaussian distribution $\hat{I}_v \sim N(I_v^*, \sigma_r^2 + \sigma_s I_v^*)$ where $\hat{I}$ is the tensor of noisy inputs, I^* is the ground truth signal, and σ_r and σ_s are noise parameters that are fixed for each sequence (Tanay et al., 2023), (Bhat et al., 2021), (Xia et al., 2020). We consider four scenarios with moderate to high gain levels [4, 8, 16, 20], corresponding to the $(\log(\sigma_r), \log(\sigma_s))$ values $[(-1.44, -1.84), (-1.08, -1.48), (-0.72, -1.12), (-0.6, -1.0)]$ respectively (Tanay et al., 2023). We evaluate all rig positions in the 10 validation scenes of the Spaces dataset for the 4 gain levels.

For all tasks, we use the L1 loss and train the model for 400k steps with the Adam optimizer, setting the learning rate to 1e-3. After 200k steps, we reduce the learning rate to 1e-4, and after 300k steps, we

Table 2

Comparison with state-of-the-art methods under the setup of 3 input views in the LLFF dataset. The best and the second best results are highlighted and underlined.

Method	Setting	3 input views		
		PSNR↑	SSIM↑	LPIPS↓
Mip-NeRF (Barron et al., 2021)	scene-specific	14.62	0.351	0.495
DietNeRF (Jain et al., 2021)		14.94	0.370	0.496
RegNeRF (Niemeyer et al., 2022)		19.08	0.587	0.336
MixNeRF (Seo et al., 2023b)		19.27	0.629	–
FlipNeRF (Seo et al., 2023a)		19.34	0.631	–
DSNeRF (Deng et al., 2022)		18.94	0.582	–
SparseNeRF (Wang et al., 2023)		19.86	0.624	0.328
SRF (Chibane et al., 2021)	fintune	12.34	0.250	0.591
PixelNeRF (Yu et al., 2021)		7.93	0.272	0.682
MVSNerF (Chen et al., 2021)		17.25	0.557	0.356
ConvGLR (Tanay & Maggioni, 2024)		<u>19.95</u>	<u>0.700</u>	0.262
Ours-S	w/o fintune	19.10	0.686	<u>0.163</u>
Ours-S-ft	fintune	20.47	0.724	0.155

further lower it to 1e-5, with a batch size of 2. We set the base channel dimension to 64 and the MLP hidden dimension to 512. To balance quality and speed, we evaluate two configurations: a lightweight Ours-S ($D=16$) and a high-fidelity Ours-L ($D=48$). As shown in Tables 1 and 5, increasing D enables finer geometric sampling, boosting PSNR from 36.08 dB to 36.29 dB, at the cost of construction time (4.00s vs. 8.40s). Thus, we recommend $D=16$ for real-time applications and $D=48$ for quality-critical tasks. We adopt an image patch size of 192×192 , and all metrics are calculated by averaging the target views of the test scenes under each setting.

5.3. Comparison methods

In order to comprehensively validate the performance of our method on novel view synthesis tasks, we conduct comparisons with a range of state-of-the-art methods, including MPI-based methods, Gaussian-based methods and NeRF-based methods:

MPI-based Methods: First we selected several of the most representative MPI-based methods for comparison, including DeepView (Flynn et al., 2019), NeX (Phongthawee et al., 2023), fMPI (Kohler et al., 2023), MPFER (Tanay et al., 2023) and ConvGLR (Tanay & Maggioni, 2024).

Gaussian-based Methods: We have also selected some of the most advanced Gaussian-based methods for comparison, including 3DGS (Kerbl et al., 2023), 2DGS (Huang et al., 2024a), DNGS (Li et al., 2024), DRGS (Chung et al., 2024), GSPro (Cheng et al., 2024), SurfelGS (Dai et al., 2024), GoF (Huang et al., 2024b), and MPGS (Li et al., 2025a).

NeRF-based Methods: Additionally, for a more comprehensive comparison, we have selected some representative NeRF-based methods, including MipNeRF (Barron et al., 2021), DietNeRF (Jain et al., 2021), RegNeRF (Niemeyer et al., 2022), MixNeRF (Seo et al., 2023b), FlipNeRF (Seo et al., 2023a), DSNeRF (Deng et al., 2022), SparseNeRF (Wang et al., 2023), SRF (Chibane et al., 2021), PixelNeRF (Yu et al., 2021), and MVSNerF (Chen et al., 2021).

At the same time, in order to comprehensively validate the performance of our method on multi-frame denoising tasks, we consider two path-based approaches: VBM4D (Maggioni et al., 2012) and VNLB (Arias & Morel, 2018), as well as four state-of-the-art learning-based methods: BPN (Xia et al., 2020), BasicVSR (Chan et al., 2021) and its extension BasicVSR++ (Chan et al., 2022), DeepRep (Bhat et al., 2021) and MPFER (Tanay et al., 2023).

5.4. Comparison on spaces datasets

In this section, we compare the proposed method with state-of-the-art approaches under both dense and sparse input conditions and present

Table 3

Comparison with state-of-the-art gaussian splatting methods under the setup of 15 input views in the space dataset. The best and the second best results are highlighted and underlined.

Method	15 input views			
	PSNR↑	SSIM↑	LPIPS↓	Params↓
3DGS (Kerbl et al., 2023)	26.96	0.869	0.228	34.5
2DGS (Huang et al., 2024a)	28.44	0.897	0.182	46.5
DRGS (Chung et al., 2024)	27.54	0.872	0.214	361.3
DNGS (Li et al., 2024)	27.79	0.898	0.214	161.0
GSPro (Cheng et al., 2024)	27.41	0.873	0.164	63.3
SurfelGS (Dai et al., 2024)	27.46	0.887	0.189	73.7
GoF (Huang et al., 2024b)	28.78	0.899	0.164	46.9
MPGS (Li et al., 2025a)	<u>29.07</u>	<u>0.907</u>	<u>0.155</u>	<u>27.9</u>
Ours-S	29.19	0.956	0.136	13.0

both qualitative and quantitative results. The common compared Spaces dataset (10 scenes) (Flynn et al., 2019) are used as the dataset.

5.4.1. Dense input task

Previous MPI-based methods often fix the input and target views during training and testing. To further verify the robustness of our method to perspective changes without affecting the comparison with existing MPI methods, we designed two experimental setups. First, to enable comparison with most existing MPI-based methods, we selected a fixed set of 12 input views (Flynn et al., 2019; Penner & Zhang, 2017; Phongthawee et al., 2023; Tanay et al., 2023; Tanay & Maggioni, 2024) and used the remaining 4 views as output for training. We crop a 16-pixel boundary across all images and average the results, as done in Flynn et al. (2019), Tanay et al. (2023), Tanay and Maggioni (2024). The numerical results are illustrated in part of Table 1, demonstrate the ours-L and ours-S models achieve the best and second-best performance, respectively. Moreover, our method offers a clear advantage in terms of parameter count and the required number of depth planes. Specifically, ours-S model uses only 2.1% of the parameters required by NeX (Phongthawee et al., 2023), and just 8.3% and 12.5% of the depth planes compared to NeX (Phongthawee et al., 2023) and ConvGLR (Tanay & Maggioni, 2024), respectively. Notably, our method generalizes well to unseen scenes and imposes no restrictions on the number of output views or their pose range, offering significantly greater flexibility. The visual results are shown in Fig. 4. To provide a more intuitive comparison of the synthesized views, we also present the corresponding error maps. It is evident that our method achieves better view quality in regions with depth discontinuities, and the reconstructed views show clearer details compared to other methods, even in complex backgrounds and occluded areas.

Next, to verify the robustness to view changes, we randomly selected 15 out of 16 views as inputs and used the remaining one as the output for training. During testing, we set each of these 16 views as the target view for testing and the final test result is obtained by averaging the performance across all 16 views without any boundary-cropping. As shown in Table 3, we compared our method with Gaussian-based methods and find that it outperforms other approaches in terms of PSNR, SSIM and LPIPS, while requiring the fewest parameters.

5.4.2. Sparse input task

To evaluate the performance of our method under sparse input conditions, we followed previous works (Flynn et al., 2019) and conducted three experimental settings, each using 4 input views. These settings were categorized as small baseline, medium baseline and large baseline according to the distances between the input views. For each setting, the number of output views was set to 3, 6 and 9, respectively. As shown in part of Table 1, our 48 layer model still outperforms all other methods and achieves the best results, while the 16 layer model maintains the second-best performance in terms of SSIM. Compared to the dense

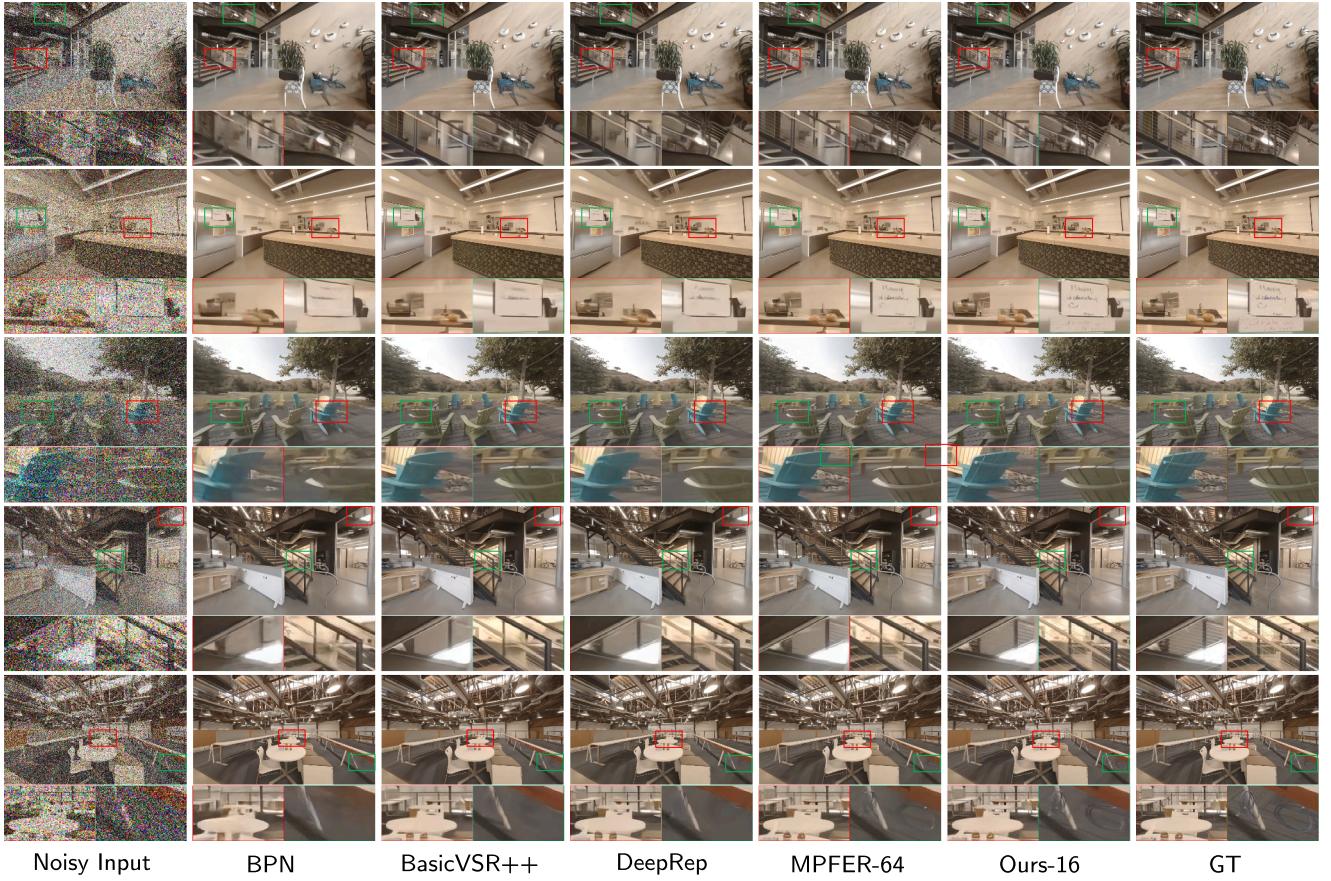


Fig. 6. Qualitative comparisons for multi-frame denoising with Gain 20 on Spaces Dataset (Flynn et al., 2019).

Table 4

Denoising on spaces. All metrics were computed on frame 6, as BPN (Xia et al., 2020) and deeprep (Bhat et al., 2021) are burst-denoising methods producing only one output.

Method	Gain 4			Gain 8			Gain 16			Gain 20			GFlops@500×800
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	
VBM4D (Maggioni et al., 2012)	32.30	0.900	–	30.12	0.849	–	27.53	0.763	–	26.58	0.723	–	–
VNLB (Arias & Morel, 2018)	33.41	0.917	0.089	30.31	0.869	0.144	25.79	0.794	0.279	23.58	0.746	0.363	–
BPN* (Xia et al., 2020)	34.52	0.934	0.048	32.10	0.900	0.082	29.45	0.846	0.144	28.56	0.824	0.168	<u>810</u>
BasicVSR (Chan et al., 2021)	36.86	0.957	0.029	34.45	0.935	0.052	31.62	0.895	0.099	30.59	0.875	0.124	2090
BasicVSR++ (Chan et al., 2022)	36.81	0.957	0.030	34.39	0.934	0.051	31.62	0.895	0.091	30.60	0.875	0.111	4300
DeepRep* (Bhat et al., 2021)	37.37	0.963	0.024	35.13	0.943	0.043	32.37	0.906	0.085	31.32	0.888	0.107	3230
MPFER-16 (Tanay et al., 2023)	37.20	0.965	0.021	35.37	0.952	0.033	33.22	0.927	0.055	32.41	0.915	0.067	470
MPFER-32 (Tanay et al., 2023)	37.52	0.967	0.020	35.69	0.954	0.030	33.50	0.931	0.051	32.66	0.919	0.063	1210
MPFER-64 (Tanay et al., 2023)	37.60	0.968	0.020	35.78	0.955	0.030	33.58	0.932	0.050	32.74	0.920	0.061	1810
Ours-S	39.15	0.986	0.013	35.96	0.975	0.017	33.89	0.949	0.025	33.59	0.937	0.028	1335

input settings, the sparse input task is more challenging. Nevertheless, our method consistently achieves strong performance while maintaining extremely low parameter counts and a small number of depth planes, demonstrating the superiority of our proposed framework in terms of efficiency and effectiveness. The visual results are shown in Fig. 5, where we compare the synthesized outputs under the most challenging large baseline setting among the three configurations. Key regions are highlighted with red boxes. As can be seen, other methods fail to effectively maintain cross-depth consistency within the MPI representation, leading to artifacts or misalignments in areas with significant depth variation, such as object boundaries. In contrast, our method demonstrates superior performance in these regions.

5.5. Comparison on RFF datasets

To evaluate the generalization of our method, we use the RFF (Mildenhall et al., 2020b) dataset as the test set in this section and compare our approach with NeRF-based methods, using only 3 input views. As shown in Table 2, compared to NeRF-based methods, our approach can directly generalize to unseen scenes. Specifically, we use the Spaces dataset for training and evaluate on the test scenes from the RFF (Mildenhall et al., 2020b) dataset without any finetuning. Even under this zero-shot setting, our method outperforms 60% of existing methods. Subsequently, after finetuning the trained model on the RFF (Mildenhall et al., 2020b) dataset, we observe further improvements,

Table 5

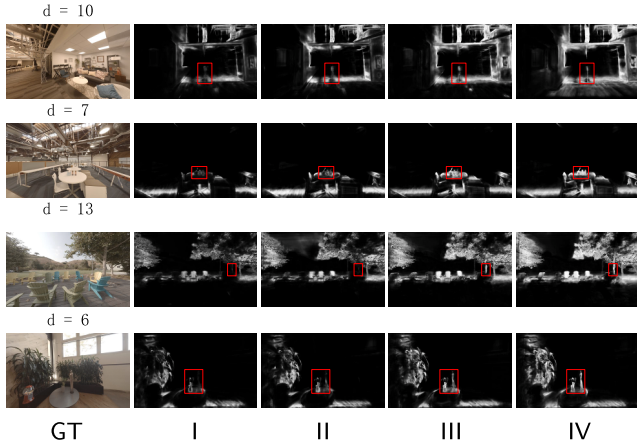
Comparison of computational efficiency under the setup of 4 input views in the spaces dataset.

Methods		Scene Representation	D	Params	Construction Time↓	Rendering Speed (FPS)↑
Scene-specific	NeX (Phongthawee et al., 2023)	NeX	192	604M	–	268.0
	MPGS (Li et al., 2025a)	3DGS	16	37.2M	–	483.0
General-izable	ConvGLR (Tanay & Maggioni, 2024)	–	128	40.3M	–	1.4
	SIMPLI (Solovev et al., 2022)	MPLI	8	2.0M	2.96s	309.6
	DeepView (Flynn et al., 2019)	MPI	80	1.7M	50.00s	20.4
	fMPI-S (Kohler et al., 2023)		16	–	0.034s	25.0
	MPFER (Tanay et al., 2023)		16	–	0.48s	1.1
	Ours-S		16	13M	4.00s	63.9
	Ours-L		48	16M	8.40s	38.9

Table 6

Quantitative comparisons for ablation study.

	F_G	$\mathcal{M}_{il}$	$\mathcal{M}_{iv}$	PSNR↑	SSIM↑
I				35.44	0.963
II	✓			35.67	0.970
III	✓	✓		35.86	0.980
IV	✓	✓	✓	36.08	0.986

**Fig. 7.** Qualitative comparisons for ablation study.

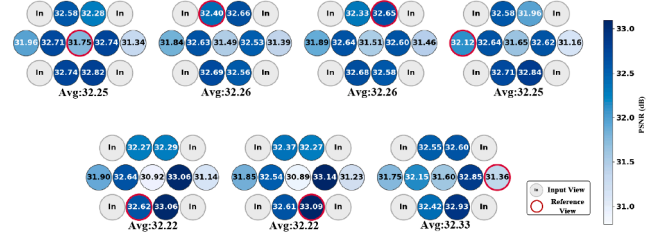
with our method achieving the best performance across all metrics, including PSNR, SSIM and LPIPS.

5.6. Comparison on multi-frame denoising

We evaluated the first equipment position of 10 validation scenes in the Spaces dataset at four gain levels without boundary cropping, with the quantitative results shown in Table 4. Each model receives 16 noisy images as input and generates 16 denoised clean images as output, where BPN and DeepRep generate only one output. For these methods, we selected the sixth frame as the output and compared the performance of all methods on that frame. As shown in Table 4, our method significantly outperforms the other methods on all metrics, and the computational cost is much reduced compared to the standard MPI methods, as we only use 16 depth planes. In Fig. 6, we also present a comparison of the qualitative results between our method and other methods. It is evident that our method produces cleaner images.

5.7. Efficiency analysis

As shown in Table 5, we present a quantitative comparison of MPI construction time and rendering speed (FPS). The results demonstrate that our method achieves a superior trade-off between construction efficiency and rendering speed. Regarding construction time, despite em-

**Fig. 8.** Per-view PSNR distribution under different reference view configurations.

ploying a serial, layer-by-layer mechanism to ensure geometric consistency, our method achieves a competitive runtime of only 4.00s. This is significantly more efficient than DeepView. In terms of rendering, our method achieves a real-time performance of 63.9 FPS. While scene-specific methods such as NeX, MPGS, and sparse representations like SIMPLI can achieve higher frame rates, they either require time-consuming scene-by-scene optimization or rely on simplified geometry layers. Crucially, among standard MPI-based general-purpose methods, our method significantly outperforms existing state-of-the-art baseline methods. Specifically, our rendering speed is approximately 2.5 times faster than fMPI-S and approximately 3 times faster than DeepView. This validates that our parallel rendering scheme maximizes efficiency without being limited by the bottleneck of the serial construction process.

5.8. Ablation studies

We compare our model with several variants to prove the effectiveness of our method. To ensure a fair comparison, we control the number of channels and depth planes so that the parameter counts of the variant models are identical.

We first remove all of our proposed modules, leaving only the standard MPI construction pipeline (I). Table 6 shows that this model suffers from significant performance degradation, with a PSNR drop of over 0.6db compared to our model. The α visualization in Fig. 7 also shows a significant loss in foreground occlusion compared to our model. Then based on (I), we add our proposed global feature F_G , sharing it directly across each layer without filtering (II). It slightly improves performance by capturing the overall scene geometry, but interference from redundant information still limits its effectiveness. Compared to I, it constructs the α structure, but artifacts are still noticeable around object edges. Next we add the Inter-layer Consistency Mask $\mathcal{M}_{il}$ (III) to filter out redundant information contained in the global feature. Compared to (II), this model further improves performance and can more clearly restore the alpha structure. It also better handles artifacts at object edges, effectively ensuring cross-layer consistency of MPI. Finally, we add Inter-view Consistency Mask $\mathcal{M}_{iv}$ (IV) to specifically address occlusion relationships in the scene. Compared to (III), the PSNR of this model is

improved by 0.2db. As can be clearly seen in Fig. 7, this mask enables the model to more accurately restore the scene's geometric structure and effectively identify foreground occlusion.

To evaluate the robustness of our proposed method to suboptimal frustum alignment, we performed a sensitivity analysis on the Spaces dataset, testing seven reference view configurations from centered to asymmetric with the input parameters fixed. As shown in Fig. 8, the method exhibits excellent stability, with the average PSNR fluctuating minimally between 32.22 dB and 32.33 dB. This statistical invariance confirms that our serial construction mechanism effectively decouples geometry construction from texture aggregation, where a consistent mask ensures accurate handling of occlusion regardless of the reference frame. Therefore, the synthesis quality depends primarily on the source-target distance, rather than the frustum alignment, indicating that our method does not overfit to specific symmetric configurations.

6. Conclusion and discussion

We propose a novel consistency-guided MPI construction method for novel view synthesis. By integrating cross-view and cross-layer consistency masks, our approach effectively handles foreground occlusions and incorporates depth-wise contextual information. Quantitative and qualitative experiments demonstrate that our method outperforms SOTA methods by 0.6%–1.1% in view synthesis and 2.1% in denoising, verifying its robustness and generalization ability in real-world scenes.

Like most MPI-based representations, our method relies on a pre-defined number of depth planes to discretize the scene. This fixed sampling may lead to discretization artifacts when representing continuous surfaces with large depth variations. In future work, we aim to investigate adaptive depth sampling techniques to dynamically adjust plane distribution, thereby reducing redundancy and further optimizing both storage and rendering efficiency for mobile deployment.

CRedit authorship contribution statement

Yichang Lv: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization; **Shuo Zhang:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Formal analysis, Conceptualization; **Jiajun Chen:** Writing – review & editing, Validation; **Chen Gao:** Writing – review & editing, Validation; **Youfang Lin:** Writing – review & editing, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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